

Research Proposal

Beyond Optical PPG: mmWave Radar Arterial Pulse Sensing

Principal Investigator: Dr. Sampath Perera, Senior Lecturer, University of Moratuwa

Co-Investigators:

- Dr. Joshua Pranjeevan Kulasingham (Senior Lecturer, University of Moratuwa)
- Thivya Kandappu (Assistant Professor, Singapore Management University)

1. Introduction

Continuous, non-invasive monitoring of cardiovascular signals has become a cornerstone of modern health and fitness technology, with photoplethysmography (PPG) being the dominant sensing modality in wearable devices. Optical PPG estimates physiological parameters such as heart rate and pulse waveform by measuring light absorption changes caused by blood volume variations in peripheral tissue. Despite its widespread adoption, optical PPG suffers from well-documented limitations: it is highly susceptible to motion artifacts [1, 2], sensitive to ambient light interference [1], affected by sweat at the skin sensor interface [3], and its accuracy varies with skin pigmentation [2] and subcutaneous fat thickness [1].

This project investigates an alternative approach to physiological sensing based on FMCW mmWave radar. mmWave sensing is an evolving modality that has been explored both as a wearable [4] and contactless distance based sensing [5]. Rather than relying on optical absorption, mmWave sensing detects the mechanical deformation of tissue caused by arterial pulsation, offering a route to physiological measurement that is inherently robust to the optical and skin related factors that degrade PPG.

In most existing work, however, data are collected in static environments, and heart rate and blood pressure are analysed independently. Some literature has shown that the pulse wave morphology obtained from mmWave sensing is closer to the ground truth arterial pulse than that from optical PPG [4]. It is also believed that radio wave sensing is affected less by skin tone and skin fat layer than optical methods, although this has not yet been well proven.

The project concentrates its effort on the signal processing and sensing aspects of the system. In particular, it explores how FMCW mmWave signals can be exploited to extract physiological signals and enable accurate estimation of biometrics such as heart rate, heart rate variability and blood pressure [6, 7]. To this end, the project aims to prove measurement accuracy across skin tone and skin thickness, to generalise the algorithm for heart rate and blood pressure, and to take radar sensing one step further by making it adaptive: unlike existing radar systems with a fixed pipeline, the SNR of the received signal may vary during motion, so the proposed system adjusts its processing accordingly.

The outcome of the project is an adaptive functional wearable sensing prototype that integrates a 60 GHz mmWave adaptive radar, an optical PPG front-end, and an inertial measurement unit (IMU), together with a comprehensive experimental comparison of mmWave and optical sensing [4]. By validating the prototype against clinical grade references a Polar H10 chest strap for heart rate and an Omron BP4350 monitor for blood pressure across a range of activity levels and participant characteristics, this work aims to contribute toward next generation cardiovascular monitoring systems that operate reliably in demanding real world environments.

2. Objectives

2.1 Primary Objectives

The primary aim of this project is to develop and evaluate an adaptive non invasive physiological sensing system based on FMCW mmWave radar that overcomes the core limitations of optical PPG. Specifically, the project seeks to:

- Develop a wearable adaptive mmWave based physiological sensing prototype using an existing commercial radar sensor capable of continuously acquiring RF signals for pulse waveform monitoring under both stationary and active conditions such as sitting, walking, and running. [4].
- Design a signal processing pipeline that exploits the FMCW mmWave signal to extract pulse wave features and estimate cardiovascular parameters [5].
- Adapt the output signal quality of the prototype based on measured parameters such as signal to noise ratio (SNR)
- Accurately estimate heart rate, blood pressure and pulse waveform characteristics from the reconstructed mmWave signals.
- Demonstrate improved robustness and accuracy across different skin pigmentation levels and subcutaneous fat thicknesses compared to optical PPG.

2.2 Secondary Objectives

Supporting the primary aim, the project also pursues the following secondary objectives:

- Conduct a comparative experimental evaluation of mmWave-based and optical PPG-based sensing across a range of activity levels (sitting, walking and running), benchmarked against a Polar H10 [8] and Omron BP4350 [9] reference devices.
- Integrate radar, optical PPG, and IMU data within a single multimodal prototype to enable synchronised, comparative signal acquisition.
- Compile a labelled physiological dataset collected from the developed prototype to support signal quality assessment and future extensions of the research.

3. Methodology

3.1 Participant Details

The study will recruit approximately 30 healthy adult volunteers ranging in age from 22 to 26 years (University of Moratuwa students). However, if sufficient variation in skin tone and BMI/subcutaneous fat thickness cannot be achieved within the UoM student population, participants may also be recruited from outside the university. Participation will be entirely voluntary, and written informed consent will be obtained from every participant prior to data collection. Participants will be selected to represent a range of skin pigmentation levels and subcutaneous fat thicknesses, as these factors are directly relevant to evaluating the performance of the sensing modalities.

Inclusion criteria require that participants be healthy individuals within the stated age range who are capable of providing informed consent. Individuals with diagnosed cardiovascular diseases or conditions affecting peripheral circulation will be excluded, since such conditions could confound the physiological measurements.

3.2 Study Location and Setting

All data collection will be conducted within the premises of the University of Moratuwa. Participants will be drawn primarily from the university community and, where necessary for diversity in skin tone and body fat category, from outside the University. The experimental environment will accommodate the range of activity conditions under investigation, allowing controlled acquisition of physiological signals during sitting, walking, and running.

3.3 Recruitment Procedure

Participants will be recruited using convenience sampling, primarily from the University of Moratuwa community and, where necessary to capture a wider range of skin tones and body fat categories, from outside the University. Interested volunteers will be provided with a study information sheet describing the purpose, procedure, and scope of the study, and will then be screened against the inclusion and exclusion criteria for eligibility. Eligible volunteers who agree to participate will provide written informed consent before any data is collected. The information sheet and consent form are used to ensure that participation is entirely voluntary and free from any dependent relationship pressure. No compensation will be offered.

3.4 Experimental Procedure

Each participant will take part in a single data collection session lasting approximately 60–90 minutes, conducted in six stages.

Basic demographic and anthropometric information age, sex, height, and weight will first be recorded for analysis purposes. Skin tone will be assessed using the Fitzpatrick classification, and body fat category will be derived from the body mass index (BMI).

The participant will then be fitted with three devices: a custom multimodal wristband prototype, a Polar H10 chest strap serving as the heart rate reference [10], and an Omron BP4350 wrist monitor serving as the blood pressure reference [9]. The wristband prototype and the Omron BP4350 will be placed on opposite wrists, both positioned over the radial artery, and the Polar H10 will be fitted around the chest. The wristband integrates a 60 GHz mmWave radar module, an optical PPG sensor, and an inertial measurement unit (IMU). Signal quality will be verified for all sensors before recording begins.

A baseline recording of approximately 5–10 minutes will then be taken with the participant seated at rest, providing a stable reference condition for later comparison. Following the baseline, the participant will perform a sequence of physical activities sitting (rest, extended duration), walking (self paced, on a flat indoor surface), and running (light jogging pace, on a treadmill or indoor track) while physiological signals are continuously and simultaneously recorded from the wristband prototype and the reference devices. Short rest intervals of about 2–3 minutes will be provided between activities to allow the heart rate to stabilise and to reduce participant fatigue. After the protocol, all devices will be removed and the participant debriefed.

This protocol allows comparison of radar based and optical PPG based heart rate and blood pressure monitoring across varying activity levels and evaluation of participant specific factors (skin tone, subcutaneous fat thickness) and physiological conditions such as sweating on sensing performance. All procedures are non invasive and use wearable, contact based sensors only; no blood, tissue, or other invasive samples are collected at any stage.

3.5 Data Analysis

The acquired signals will first be preprocessed to remove noise and apply appropriate filtering. Relevant features, including heart rate and blood pressure, will then be extracted using signal processing algorithms [5]. The pulse waveforms, estimated heart rate, and estimated blood pressure from the wristband prototype will be validated against synchronized reference measure-

ments from the Polar H10 (heart rate) and the Omron BP4350 (blood pressure), using accuracy and reliability metrics. This comparative analysis will quantify the performance of mmWave based sensing relative to optical PPG, and will assess how participant specific and physiological factors influence each modality.

4. Ethical Considerations

4.1 Risks and Mitigation

This study is classified as minimal risk, as it relies entirely on non-invasive sensing. Participants come into contact with a wearable wristband prototype, Polar H10 chest strap and Omron BP4350 wrist monitor. The radar transmits only a electromagnetic wave toward the tissue to measure mechanical deformation from arterial pulsation; this constitutes safe, low-power non-ionizing radiation that is standard for commercial near-field sensing, and no electrical current is applied to the participant. All sensing components are used within their intended manufacturer specifications and are of a type widely adopted in prior physiological monitoring research. All sensing components used here have already been used in previous human based research studies, and relevant references are provided for each component [4] [10] [9] . The physical activities involved (sitting, walking, and running) carry no more risk than everyday exertion; participants with diagnosed cardiovascular disease or conditions affecting peripheral circulation are excluded to avoid exposing at-risk individuals to activity-related strain. Participation is voluntary, and participants may withdraw at any time without consequence.

4.2 Benefits

Participants may not receive any direct medical benefit from taking part in the study, though they may gain awareness of their own physiological measurements recorded during the session. The broader benefit is to the wider community: the study contributes to the development of robust, non-invasive cardiovascular monitoring systems that address the limitations of conventional optical sensors including motion artifacts, ambient light interference, skin pigmentation effects, and subcutaneous tissue variation thereby supporting the advancement of next-generation health monitoring technology.

4.3 Consent

Written informed consent will be obtained from every participant before any data is collected. Prior to consent, each volunteer will be given an information sheet explaining the purpose, procedures, scope, and voluntary nature of the study. The information sheet and consent form will also be provided in the participants' first language. The information sheet and consent form are used specifically to ensure that participation is purely voluntary and free from any dependent-relationship pressure. All participants are competent adults capable of giving voluntary and informed consent, and each is free to withdraw at any point.

4.4 Confidentiality and Data Security

All collected data will be handled in strict confidence. Electronic data files will be labelled using anonymous identification codes rather than personal identifiers, and consent forms will be stored separately from the research data. During the study, paper documents will be kept in a locked cabinet within the Department of Electronic and Telecommunication Engineering, while electronic data will be stored on password-protected computers accessible only to the research team. Any results appearing in publications, theses, or conference presentations will be reported only in anonymised form. Data may be preserved for future related research only with additional participant consent, with access limited to the research team and approved collaborators.

5. Annex A — The Multimodal Physiological Sensing System

5.1 General Overview

The sensing system developed for this project consists of a custom multimodal wristband prototype together with commercial reference devices. The wristband integrates three sensing modalities: a 60 GHz mmWave radar module, which detects the mechanical deformation of tissue caused by arterial pulsation; an optical photoplethysmography (PPG) front-end, which measures blood-volume changes through light absorption; and an inertial measurement unit (IMU), which captures body motion to support signal processing. The wristband is worn on the wrist, positioned over the radial artery, and acquires RF, optical, and inertial signals simultaneously. A Polar H10 chest strap serves as the ground-truth reference device for heart rate and Omron BP4350 wrist monitor serving as the blood pressure reference. The acquired signals are transferred to a computer, where preprocessing and pulse-waveform reconstruction are performed [5]. The pipeline is designed to be adaptive: signal quality parameters such as the SNR are monitored and used to adjust the output accordingly. Together, these components form a complete pipeline of multimodal signal acquisition, synchronised reference measurement, and signal processing for physiological monitoring.

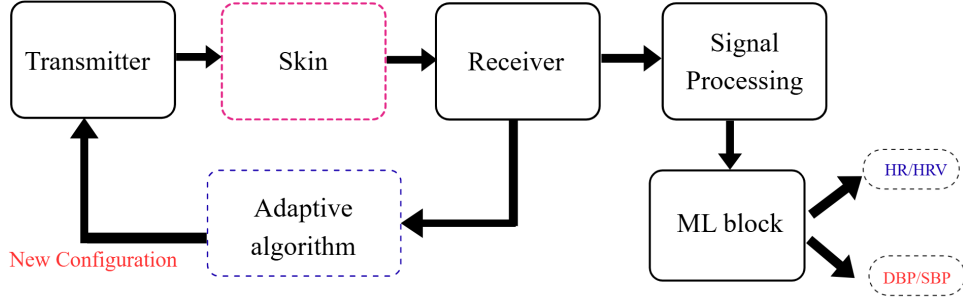


Figure 1: System Overview

5.2 Electrical Safety

All devices worn by participants are low-voltage, battery-powered units with no mains connection during data collection, eliminating any risk of electric shock from the supply. No electrical current is applied to the participant; the sensors operate purely by measuring optical, electromagnetic, and inertial signals at the skin surface. The wristband electronics are enclosed within the prototype housing so that participants do not contact exposed conductors or circuitry.

5.3 EMF Safety

The 60 GHz mmWave radar operates as a low-power near-field sensing device, transmitting only a low-power electromagnetic wave toward the tissue to measure the mechanical deformation caused by arterial pulsation. This is non-ionizing radiation and falls well within the safe exposure limits established for commercial short-range radar sensors. Participants are not exposed to ionizing radiation, and the emitted power is far below levels associated with any thermal or biological effect. The optical PPG front-end likewise uses low-intensity LED illumination consistent with standard commercial pulse-oximetry devices. Because all EMF and optical emissions are low-power and non-ionizing, no external radiation-safety approval is required.

5.4 Materials Safety

All materials in direct contact with the participant’s skin are limited to the wristband housing, strap, and the chest strap electrode band. These materials are of the type used in commercial wearable devices and are selected to be skin safe and non irritant. The wristband is worn

externally over the wrist for the short duration of each session, while the Polar H10 chest strap and the Omron BP4350 cuff are certified products designed for direct skin contact. No adhesives, chemicals, or substances that could cause an allergic or irritant reaction are applied to the skin.

5.5 Benchmarking Procedure

The performance of the prototype will be benchmarked against the Polar H10 chest strap, the ground-truth reference for heart rate, and the Omron BP4350 cuff, the reference for blood pressure. Physiological signals will be acquired simultaneously from the wristband prototype and the reference devices across the full range of activity conditions, ensuring time synchronised comparison. The heart rate, pulse-waveform characteristics, and blood pressure estimated from the mmWave and optical PPG modalities will be evaluated against the reference measurements using accuracy and reliability metrics. This benchmarking allows quantitative assessment of each sensing modality’s robustness and of how participant-specific factors such as skin pigmentation and subcutaneous fat thickness affect measurement accuracy.

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